

An Exact Algebraic Certificate for the Raigorodskii Lower-Bound Constant

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Abstract

Raigorodskii's asymptotic lower bound for the chromatic number of high-dimensional Euclidean space involves the numerical base $1.2395667407\dots$. Starting from Näslund's one-variable reformulation of this base, we give a finite exact-arithmetic certificate. We prove that the relevant rational function has a unique maximizer $s_* \in (0, 1)$, eliminate s_* , and obtain the primitive irreducible integral polynomial

$$144x^6 - 528x^5 + 648x^4 - 220x^3 - 71x^2 + 42x - 31.$$

Consequently the constant is algebraic of degree six. It is the unique real root of this polynomial greater than 1.2, and exact rational sign evaluations give a certified enclosure of width 10^{-40} . An elementary Bernstein-basis certificate, independently cross-checked by a Sturm computation, proves that the sextic has exactly two real roots. We also prove that its Galois group is S_6 , so it is not expressible by radicals, and that $6\gamma_{\mathbb{R}}$ is an algebraic integer with minimal positive integral scaling factor 6. This note does not improve the chromatic lower bound; it makes its numerical base algebraically explicit and independently reproducible.

1 Introduction and source of the optimization

Let $\chi(\mathbb{R}^n)$ denote the chromatic number of the unit-distance graph on $\mathbb{R}^n$. Raigorodskii proved

$$\chi(\mathbb{R}^n) \geq (\gamma_{\mathbb{R}} + o(1))^n,$$

where $\gamma_{\mathbb{R}} = 1.23956674\dots$ is the best known asymptotic lower-bound base for this problem [1, 2, 3, 4]. Here $\gamma_{\mathbb{R}}$ denotes the base in that lower bound, not the unknown asymptotic growth rate of $\chi(\mathbb{R}^n)$.

The existence of

$$\lim_{n \rightarrow \infty} \chi(\mathbb{R}^n)^{1/n}$$

is asked in Erdős Problem 704 [4]. The present note does not address the existence or value of that limit. It certifies exactly the known lower-bound base on one side of the current exponential corridor.

Näslund's Theorem 3 gives, for $m = k = 1$ and $\ell = 3$,

$$\max_{0 < t < 1} \frac{\theta(t^{1/2}; 3)}{1 + t + t^2} = \max_{0 < t < 1} \frac{1 + \sqrt{t} + t^{3/2}}{1 + t + t^2}.$$

After setting $s = \sqrt{t}$, define

$$\gamma_* = \max_{0 < s < 1} F(s), \quad F(s) = \frac{1 + s + s^3}{1 + s^2 + s^4}. \quad (1)$$

Näslund explicitly states that, when $k = m = 1$, the maximizing choice $\ell = 3$ retrieves Raigorodskii's lower bound. The identification of γ_* with the Raigorodskii lower-bound base is therefore inherited from Näslund; throughout the rest of the note we write $\gamma_R = \gamma_*$. The present work certifies the exact algebraic value of this specific optimum from Theorem 3. This is the sole external mathematical input to the certificate below.

Remark 1 (The display in arXiv v2). In §1.3 of arXiv:2205.12312v2, the explanatory display prints the reciprocal quotient. This cannot have the stated value: for $0 < s < 1$,

$$F(s) - 1 = \frac{s(1-s)(1+s^2)}{1+s^2+s^4} > 0,$$

so the printed reciprocal is strictly less than one. Theorem 3 gives the orientation in (1). This remark concerns the arXiv v2 display; the present draft has not verified whether the typeset journal version retained or corrected it.

2 Exact algebraic certificate

Define

$$P(s) = 1 - 2s + 2s^2 - 4s^3 - 2s^4 - s^6, \quad (2)$$

$$Q(x) = 144x^6 - 528x^5 + 648x^4 - 220x^3 - 71x^2 + 42x - 31. \quad (3)$$

Theorem 2. *Let γ_R be defined by (1). Then:*

- (i) *F has a unique maximizer $s_* \in (0, 1)$, and s_* is the unique root of P in $(0, 1)$.*
- (ii) *$Q(\gamma_R) = 0$. The polynomial Q is primitive and irreducible over $\mathbb{Q}$. Thus γ_R has degree six over $\mathbb{Q}$; its monic minimal polynomial is $Q/144$, and Q is its primitive irreducible integral associate.*
- (iii) *Q is strictly increasing on $[6/5, \infty)$, with*

$$Q\left(\frac{6}{5}\right) = -\frac{49351}{15625} < 0, \quad Q\left(\frac{13}{10}\right) = \frac{88379}{15625} > 0.$$

Consequently γ_R is the unique real root of Q greater than 1.2.

- (iv) *The exact enclosure*

$$1.2395667407265985397097751707397283370137 < \gamma_R,$$

$$\gamma_R < 1.2395667407265985397097751707397283370138$$

holds. Its width is 10^{-40} .

(v) The polynomial Q has exactly two real roots: one in $(-0.467, -0.466)$ and the positive root γ_R in $(1.2, 1.3)$.

Proof. Write

$$N(s) = 1 + s + s^3, \quad D(s) = 1 + s^2 + s^4.$$

Direct differentiation gives

$$F'(s) = \frac{N'(s)D(s) - N(s)D'(s)}{D(s)^2} = \frac{P(s)}{(1 + s^2 + s^4)^2}. \quad (4)$$

Moreover,

$$-P'(s) = 6s^5 + 8s^3 + 12s^2 - 4s + 2.$$

For $s > 0$, the first two terms are nonnegative, while $12s^2 - 4s + 2 > 0$ because its discriminant is -80 . Hence $P'(s) < 0$ on $(0, \infty)$. Since $P(0) = 1$ and $P(1) = -6$, the polynomial P has exactly one root $s_* \in (0, 1)$. Equation (4) then shows that F increases before s_* and decreases after it. This proves (i).

To eliminate s , impose

$$P(s) = 0, \quad xD(s) - N(s) = 0.$$

An exact Sylvester resultant calculation gives

$$\text{Res}_s(P(s), x(1 + s^2 + s^4) - (1 + s + s^3)) = 3Q(x). \quad (5)$$

At $x = F(s_*) = \gamma_R$, the value $s = s_*$ is a common zero of the two polynomials in s . Their resultant therefore vanishes, and (5) gives $Q(\gamma_R) = 0$.

The coefficients of Q have greatest common divisor one, so Q is primitive. Modulo 5, multiplication by the unit -1 produces the monic polynomial

$$\overline{Q}(x) = x^6 - 2x^5 + 2x^4 + x^2 - 2x + 1 \in \mathbb{F}_5[x].$$

Exact finite-field arithmetic gives

$$\begin{aligned} \gcd(\overline{Q}, x^{5^2} - x) &= 1, \\ \gcd(\overline{Q}, x^{5^3} - x) &= 1, \\ x^{5^6} - x &\equiv 0 \pmod{\overline{Q}}. \end{aligned}$$

These are Rabin's degree-six irreducibility conditions [5]. Thus $\overline{Q}$ is irreducible over $\mathbb{F}_5$, and Gauss's lemma implies that Q is irreducible over $\mathbb{Q}$. This proves (ii).

For uniqueness of the relevant real root, put $x = 6/5 + y$. Exact expansion gives

$$\begin{aligned} Q'\left(\frac{6}{5} + y\right) &= 864y^5 + 2544y^4 + \frac{11808}{5}y^3 + \frac{19788}{25}y^2 \\ &\quad + \frac{22714}{125}y + \frac{236814}{3125}. \end{aligned}$$

Every coefficient is positive, so Q is strictly increasing on $[6/5, \infty)$. The displayed exact endpoint values show that Q has one root in $(6/5, 13/10)$. Since

$$\gamma_R \geq F\left(\frac{1}{2}\right) = \frac{26}{21} > \frac{6}{5},$$

that root is γ_R , proving (iii).

For (iv), let

$$L = \frac{12395667407265985397097751707397283370137}{10^{40}},$$

$$U = \frac{12395667407265985397097751707397283370138}{10^{40}}.$$

Exact integer arithmetic gives $Q(L) < 0 < Q(U)$. Strict monotonicity on $[6/5, \infty)$ places the unique positive root in (L, U) .

Part (v) follows from the elementary positivity certificate in Proposition 3 below. The verification programs also compute an independent exact Sturm sequence with the same root count. $\square$

Proposition 3 (Elementary certificate for the two real roots). *The polynomial Q has exactly one root in $(-467/1000, -466/1000)$, exactly one root in $(6/5, 13/10)$, and no other real roots.*

Proof. For $0 \leq t \leq 1$, write

$$B_{i,n}(t) = \binom{n}{i} t^i (1-t)^{n-i}.$$

Every $B_{i,n}$ is nonnegative on $[0, 1]$, and $\sum_{i=0}^n B_{i,n}(t) = 1$.

First, for $y \geq 0$, exact expansion gives

$$Q\left(-\frac{467}{1000} - y\right) = 144y^6 + \frac{116436}{125}y^5 + \frac{29399403}{12500}y^4 + \frac{8985296567}{3125000}y^3$$

$$+ \frac{21570510479867}{12500000000}y^2 + \frac{13886577206994389}{31250000000000}y + \frac{21899765866381721}{62500000000000000}.$$

All coefficients are positive, so $Q(x) > 0$ for $x \leq -467/1000$.

Next set

$$x = -\frac{466}{1000} + \frac{833}{500}t,$$

which maps $[0, 1]$ onto $[-466/1000, 6/5]$. Exact conversion to the degree-eight Bernstein basis gives

$$-Q(x) = \sum_{i=0}^8 b_i B_{i,8}(t),$$

where

$$(b_0, \dots, b_8) = \left(\frac{90089305534379}{9765625000000000}, \frac{359044792692858729}{3906250000000000}, \frac{52870090497122397}{3906250000000000}, \right.$$

$$\frac{124620702915641}{781250000000000}, \frac{13443213954293}{390625000000}, \frac{282934224509}{6250000000},$$

$$\left. \frac{3214343901}{125000000}, \frac{118373431}{6250000}, \frac{49351}{15625} \right).$$

Every b_i is positive. Hence $Q(x) < 0$ throughout $[-466/1000, 6/5]$.

On the remaining interval put

$$x = -\frac{467}{1000} + \frac{t}{1000}.$$

A degree-five Bernstein expansion gives

$$-Q'(x) = \sum_{i=0}^5 c_i B_{i,5}(t),$$

with

$$(c_0, \dots, c_5) = \left(\frac{13886577206994389}{31250000000000}, \frac{6932503348257261}{15625000000000}, \frac{3460865785481089}{7812500000000}, \right. \\ \left. \frac{1727743314228061}{3906250000000}, \frac{862528548927789}{1953125000000}, \frac{430593559988861}{976562500000} \right).$$

Thus $-Q'(x) > 0$, so Q is strictly decreasing on $[-467/1000, -466/1000]$. Since

$$Q\left(-\frac{467}{1000}\right) = \frac{21899765866381721}{62500000000000000} > 0, \quad Q\left(-\frac{466}{1000}\right) = -\frac{90089305534379}{976562500000000} < 0,$$

continuity gives exactly one root in that interval. Part (iii) gives exactly one root in $(6/5, 13/10)$ and none on $[13/10, \infty)$. Together with the sign certificates on $(-\infty, -467/1000]$ and $[-466/1000, 6/5]$, these intervals cover $\mathbb{R}$ and prove the claim. $\square$

Corollary 4 (Galois group, radicals, and integral scaling). *The Galois group of Q over $\mathbb{Q}$ is the full symmetric group S_6 . Consequently γ_R is not expressible by radicals. Moreover, γ_R is not an algebraic integer, while $6\gamma_R$ is an algebraic integer; the integer 6 is the least positive integer d for which $d\gamma_R$ is an algebraic integer.*

Proof. Set

$$R(y) = 6^6 \frac{Q(y/6)}{144} = y^6 - 22y^5 + 162y^4 - 330y^3 - 639y^2 + 2268y - 10044.$$

The polynomial $R \in \mathbb{Z}[y]$ is monic and irreducible, and $R(6\gamma_R) = 0$. Its discriminant is

$$\text{disc}(R) = 2^{17} 3^{13} 47^3 421^3.$$

For each $p \in \{61, 107\}$, one has $p \nmid \text{disc}(R)$, which is the hypothesis of Dedekind's factorization theorem in the form used here. Moreover, $p \nmid 6 \cdot 144 \cdot 324$: in particular,

$$144 \equiv 22 \pmod{61}, \quad 144 \equiv 37 \pmod{107}$$

are units, so $Q \bmod p$ retains degree six and no factor is lost on reduction. Since

$$R(6x) = 324Q(x),$$

the invertible substitution $y = 6x$, followed by multiplication by the unit 324, preserves the factor-degree pattern. Exact factorizations give

$$Q(x) \equiv 22(x-9)(x^5 - 15x^4 + 22x^3 + 5x^2 - 22x - 2) \pmod{61},$$

where the quintic factor is irreducible over $\mathbb{F}_{61}$, and

$$Q(x) \equiv 37(x-16)(x-11)(x+25)(x+33) \cdot (x^2+x-50) \pmod{107},$$

where the quadratic factor is irreducible over $\mathbb{F}_{107}$. By Dedekind's factorization theorem, equivalently the standard factorization-cycle correspondence at primes not dividing $\text{disc}(R)$ [6], the Galois group contains a 5-cycle and a transposition. Although irreducibility modulo 5 already supplies a 6-cycle, a 6-cycle together with an unspecified transposition does not uniformly force S_6 ; the 5-cycle is used so that the transitivity argument below is independent of which transposition occurs.

Irreducibility makes the Galois group transitive. The stabilizer of the point fixed by the 5-cycle acts transitively on the remaining five points, so the full action is 2-transitive. Conjugating the transposition then produces every transposition, and the group is S_6 . Since S_6 is not solvable, γ_R cannot be expressed by radicals.

The monic minimal polynomial of γ_R is $Q/144$, which is not in $\mathbb{Z}[x]$; hence γ_R is not an algebraic integer. The displayed monic polynomial R proves that $6\gamma_R$ is an algebraic integer.

Finally, for a positive integer d , the monic minimal polynomial of $d\gamma_R$ is

$$M_d(y) = d^6 \frac{Q(y/d)}{144}.$$

If $d\gamma_R$ is an algebraic integer, then $M_d \in \mathbb{Z}[y]$. Its coefficients of y^5 and y^4 are respectively

$$-\frac{11d}{3}, \quad \frac{9d^2}{2}.$$

Therefore $3 \mid d$ and $2 \mid d$, so $6 \mid d$. Since $d = 6$ works, the least such positive integer is exactly 6. $\square$

Remark 5 (Minimal certificate primes). Among primes not dividing $\text{disc}(R)$, an exact finite scan shows that 61 is the least prime for which $R \bmod p$ has factor degrees $[1, 5]$, and 107 is the least prime for which it has factor degrees $[1, 1, 1, 1, 2]$. This minimality is not needed for the proof, but explains the choice of the two Dedekind primes.

3 Isolation of the maximizer

The maximizer can also be isolated exactly. Put

$$s_1 = \frac{464087083432496056}{10^{18}}, \quad s_2 = \frac{464087083432496057}{10^{18}}.$$

Exact evaluation gives

$$P(s_1) > 0 > P(s_2).$$

Because P is strictly decreasing,

$$0.464087083432496056 < s_* < 0.464087083432496057.$$

This is an independent rational certificate for the optimizer used in the elimination step.

4 Reproducibility and independent exact checks

The accompanying package contains two verification programs.

The first uses SymPy 1.14.0 as an exact-arithmetic engine. It checks the derivative identity and its positivity decomposition, the resultant, a lexicographic Gröbner elimination, the minimal polynomial of the exact optimizer value, the primitive content, the Rabin certificate over $\mathbb{F}_5$, both exact real-root certificates, the endpoint brackets, and the modular factorizations used in Corollary 4 [7].

The second uses only Python’s standard library. It derives the finite-field reduction from Q , implements polynomial arithmetic over finite fields, computes the full Sylvester resultant symbolically by fraction-free Bareiss elimination over $\mathbb{Z}[x]$, constructs rational Sturm sequences, verifies independently the Bernstein-basis certificate of Proposition 3, derives R and its discriminant from Q , and checks the endpoint, integral-scaling, Dedekind, and Galois-factorization certificates. Both programs also verify that 61 and 107 are the least unramified primes with the required factor patterns.

All load-bearing Python checks use explicit exceptions rather than the `assert` statement, so they remain active under `python -O`. A directed mutation harness corrupts coefficients, enclosures, Bernstein data, and cycle-prime claims and requires every corruption to be detected in normal execution, with representative corruptions repeated under optimized execution.

A Lean 4 companion, pinned to Lean and Mathlib version 4.32.1, kernel-checks the analytic and real-algebraic core and the integrality of $6\gamma_R$. It contains no placeholder proofs or added axioms. The continuous-integration workflow scans all Lean source tokens for placeholders and records `#print axioms` output for the principal theorems. Irreducibility, the global root count, minimality of the scaling factor, and the Galois argument are not claimed as fully formalized in Lean.

No floating-point computation is used in any logical step.

5 Scope and bibliographic status

This note is an exact evaluation of a known lower-bound constant. It neither improves the asymptotic lower bound for $\chi(\mathbb{R}^n)$ nor resolves Erdős Problem 704.

To the best of our knowledge, this explicit minimal polynomial and exact root isolation have not previously appeared in the literature. Searches included the exact decimal expansion, both coefficient tuples, OEIS, accessible primary texts and bibliographic records for the cited lower-bound papers, and the public Erdős-problems record. No prior appearance was found. The search was not comprehensive: in particular, MathSciNet, zbMATH, some paywalled full texts, and Russian-language literature beyond accessible records were not exhaustively searched. No unconditional claim of bibliographic priority is made pending broader specialist review.

Research provenance and AI-assistance disclosure

This note arose from a multi-model research workflow designed and coordinated by the author, combining ChatGPT, Claude, Gemini, primary mathematical sources, exact symbolic computation, and repeated adversarial checking. The algebraic derivation and initial manuscript were principally generated by AI systems. The author selected and directed the investigation, compared model

outputs, consolidated the surviving derivations, and arranged the exact verification and external review. No AI system is listed as an author. This draft is being circulated specifically for independent human assessment of correctness, relevance, priority, and attribution.

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